SCHEDULING ALGORITHM

1.FCFS

CODE :

```
#include<stdio.h>
int main(){
        int n;
        printf("Enter number of processes: ");
        scanf("%d",&n);
        int at[n],bt[n],ct[n],tat[n],wt[n];
        for(int i=0;i<n;i++){
                printf("Enter Arrival time and Burst Time for P%d: ", i+1);
                scanf("%d %d:",&at[i],&bt[i]);
        }
        float total_tat = 0, total_wt = 0;
        ct[0] = at[0]+bt[0];
        for(int i=0;i<n;i++){
                if(ct[i-1]<at[i]){
                        ct[i]=at[i]+bt[i];
                }else{
                        ct[i]=ct[i-1]+bt[i];
                }
        }
        for(int i=0;i<n;i++){
                tat[i]=ct[i]-at[i];
                wt[i] = tat[i] - bt[i];
                total_tat += tat[i];
                total_wt += wt[i];
        }
        printf("\nAverage Turnaround Time: %.2f", total_tat/n);
        printf("\nAverage Waiting Time: %.2f", total_wt/n);
        return 0;
}
[tejini@localhost scheduling]$
```

OUTPUT:

```
Activities   Terminal                    Apr 19  13:53
                          tejini@localhost:~/scheduling
[tejini@localhost scheduling]$ gcc fcfs.c
[tejini@localhost scheduling]$ ./a.out
Enter number of processes: 4
Enter Arrival time and Burst Time for P1: 0 2
Enter Arrival time and Burst Time for P2: 1 2
Enter Arrival time and Burst Time for P3: 5 3
Enter Arrival time and Burst Time for P4: 6 4

Average Turnaround Time: 3.50
Average Waiting Time: 0.75[tejini@localhost scheduling]$
```

2.SJF

CODE :

```
[tejini@localhost ~]$ ls
a        Documents  exec    fork.c  Music     shellscript  Videos
Desktop  Downloads  exec.c  memory  Pictures  videos
[tejini@localhost ~]$ mkdir scheduling
[tejini@localhost ~]$ vi sjf.c
[tejini@localhost ~]$ cat sjf.c
#include<stdio.h>
int main(){
                int n;
                printf("Enter number of processes: ");
                scanf("%d",&n);
                int at[n], bt[n], ct[n], tat[n], wt[n];
                int done[n];
                for(int i=0;i<n;i++){
                        printf("Enter Arrival Time and Burst Time for P%d: ",i+1);
                        scanf("%d %d",&at[i], &bt[i]);
                        done[i]=0;
                }
                int completed=0;
                int completionTime=0;
                float avgtat=0, avgwt=0;
                while(completed < n){
                        int idx = -1;
                        int minBT = 9999;
                        for(int i=0; i<n; i++){
                                if(at[i]<=completionTime && done[i]==0){
                                        if(bt[i]<minBT){
                                                minBT = bt[i];
                                                idx = i;
                                        }
                                }
                        }
                        if(idx != -1){
                                if(completionTime<at[idx]){
                                                completionTime = at[idx];
                                }
                                completionTime += bt[idx];
                                tat[idx] = completionTime - at[idx];
                                wt[idx] = tat[idx] - bt[idx];
                                done[idx] = 1;
                                completed++;
                        }else{
                                completionTime++;
                        }
                }
                for(int i=0;i<n;i++){
                        avgtat += tat[i];
                        avgwt += wt[i];
                }
                printf("\nAverage Turnaround Time: %.2f\n",avgtat/n);
                printf("Average Waiting Time: %.2f\n",avgwt/n);
                return 0;
}
```

OUTPUT:

```
[tejini@localhost ~]$ gcc sjf.c
[tejini@localhost ~]$ ./a.out
Enter number of processes: 4
Enter Arrival Time and Burst Time for P1: 1 3
Enter Arrival Time and Burst Time for P2: 2 4
Enter Arrival Time and Burst Time for P3: 1 2
Enter Arrival Time and Burst Time for P4: 4 4

Average Turnaround Time: 6.25
Average Waiting Time: 3.00
[tejini@localhost ~]$
```

3. ROUND ROBIN

CODE

Activities Terminal Apr 20 07:24

tejini@localhost:~ — /usr/bin/vim roundrobin.c

```c
#include<stdio.h>
int main()
{
        int n,tq,completed=0, completionTime = 0;
        printf("Enter number of processes: \n");
        scanf("%d",&n);
        printf("Enter Time Quantum: \n");
        scanf("%d",&tq);
        int at[n],bt[n],rem[n],tat[n],wt[n];
        float avgtat = 0,avgwt=0;

        for(int i=0;i<n;i++)
        {
                printf("Enter AT and BT for P%d: \n",i+1);
                scanf("%d %d",&at[i],&bt[i]);
                rem[i]=bt[i];
        }
        while(completed<n)
        {
                int idle=1;
                for(int i=0;i<n;i++)
                {
                        if(at[i]<=completionTime && rem[i]>0)
                        {
                                idle=0;
                                if(rem[i]>tq)
                                {
                                        completionTime += tq;
                                        rem[i]-=tq;
                                }
                                else
                                {
                                        completionTime+=rem[i];
                                        tat[i]=completionTime - at[i];
                                        wt[i] = tat[i]-bt[i];
                                        rem[i]=0;
                                        completed++;
                                }
                        }
                }
                if (idle)
```

1,17 Top

Activities Terminal Apr 20 07:25

tejini@localhost:~ — /usr/bin/vim roundrobin.c

```c
                printf("Enter AT and BT for P%d: \n",i+1);
                scanf("%d %d",&at[i],&bt[i]);
                rem[i]=bt[i];
        }
        while(completed<n)
        {
                int idle=1;
                for(int i=0;i<n;i++)
                {
                        if(at[i]<=completionTime && rem[i]>0)
                        {
                                idle=0;
                                if(rem[i]>tq)
                                {
                                        completionTime += tq;
                                        rem[i]-=tq;
                                }
                                else
                                {
                                        completionTime+=rem[i];
                                        tat[i]=completionTime - at[i];
                                        wt[i] = tat[i]-bt[i];
                                        rem[i]=0;
                                        completed++;
                                }
                        }
                }
                if (idle)
                   completionTime++;
        }

        for(int i=0;i<n;i++)
        {
                avgtat += tat[i];
                avgwt += wt[i];
        }
        printf("\nAverage Turnaround Time: %.2f\n",avgtat/n);
        printf("Average Waiting Time: %.2f\n",avgwt/n);

        return 0;
}
```

54,1 Bot

OUTPUT :

```
Activities   Terminal                    Apr 20  07:26
                                   tejini@localhost:~
[tejini@localhost ~]$ vi roundrobin.c
[tejini@localhost ~]$ gcc roundrobin.c
[tejini@localhost ~]$ ./a.out
Enter number of processes:
3
Enter Time Quantum:
2
Enter AT and BT for P1:
0 5
Enter AT and BT for P2:
1 3
Enter AT and BT for P3:
2 1

Average Turnaround Time: 6.33
Average Waiting Time: 3.33
[tejini@localhost ~]$
```

4. PRIORITY SCHEDULING

CODE

```
Activities   Terminal                    Apr 20  07:38
                   tejini@localhost:~/scheduling — /usr/bin/vim priority.c
#include<stdio.h>
int main(){
        int n,completed=0,time=0;
        printf("Enter number of processes: \n");
        scanf("%d",&n);
        int at[n],bt[n],priority[n],done[n];
        int ct[n],tat[n],wt[n];
        float avgtat =0,avgwt=0;

        for(int i=0;i<n;i++)
        {
                printf("Enter AT,BT and priority for P%d: \n",i+1);
                scanf("%d %d %d",&at[i],&bt[i],&priority[i]);
                done[i]=0;
        }
        while(completed < n)
        {
                int idx=-1;
                int highPriority = 9999;
                for(int i=0;i<n;i++)
                {
                        if(at[i] <=time && done[i]==0)
                        {
                                if(priority[i] < highPriority)
                                {
                                        highPriority = priority[i];
                                        idx=i;
                                }
                        }
                }
                if(idx!=-1)
                {
                        time += bt[idx];
                        ct[idx]=time;
                        tat[idx] = ct[idx] - at[idx];
                        wt[idx] = tat[idx] - bt[idx];
                        done[idx] = 1;
                        completed++;
                }
                else
                {
"priority.c" 53L, 952B                                          3,19-26       Top
```

Activities Terminal Apr 20 07:39

tejini@localhost:~/scheduling — /usr/bin/vim priority.c

```
                scanf("%d %d %d",&at[i],&bt[i],&priority[i]);
                done[i]=0;
        }
        while(completed < n)
        {
                int idx=-1;
                int highPriority = 9999;
                for(int i=0;i<n;i++)
                {
                        if(at[i] <=time && done[i]==0)
                        {
                                if(priority[i] < highPriority)
                                {
                                        highPriority = priority[i];
                                        idx=i;
                                }
                        }
                }
                if(idx!=-1)
                {
                        time += bt[idx];
                        ct[idx]=time;
                        tat[idx] = ct[idx] - at[idx];
                        wt[idx] = tat[idx] - bt[idx];
                        done[idx] = 1;
                        completed++;
                }
                else
                {
                        time++;
                }
        }
        for(int i=0;i<n;i++)
        {
                avgtat +=tat[i];
                avgwt +=wt[i];
        }
        printf("\nAverage Turnaround time: %.2f\n",avgtat/n);
        printf("Average Waiting Time: %.2f\n", avgwt/n);
        return 0;
}
                                                                53,1          Bot
```

OUTPUT :

Activities Terminal Apr 20 07:26

tejini@localhost:~

```
[tejini@localhost ~]$ vi roundrobin.c
[tejini@localhost ~]$ gcc roundrobin.c
[tejini@localhost ~]$ ./a.out
Enter number of processes:
3
Enter Time Quantum:
2
Enter AT and BT for P1:
0 5
Enter AT and BT for P2:
1 3
Enter AT and BT for P3:
2 1

Average Turnaround Time: 6.33
Average Waiting Time: 3.33
[tejini@localhost ~]$
```